

Apoorva Gupta

Junior Undergraduate
Department of Computer Science and Engineering

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech CSE	Indian Institute of Technology Kanpur	9.63/10
2021	CBSE (XII)	Bharat Ram Global School, Greater Noida	97.4%
2019	CBSE (X)	Somerville School, Greater Noida	97.2%

Scholastic Achievements

- Awarded with **Dr. Sangeeta Goel Memorial Award** for the **best female** 1st year undergraduate in the year 2021-2022
- Selected for **Google Girl Hackathon'23** from IIT Kanpur, attended Learning Cohort Workshops taken by experts at google
- Achieved a maximum **codeforces** rating of **1560+** with rank **773**(CF-878 Div 2) and **793**(ECF-149 Div 2) on ap234gupta
- Received **Academic Excellence Award** by IIT Kanpur for exceptional academic performance twice in the session 2021-22
- Received **KVPY (SX) fellowship** in year 2020 issued by the Department of Science and Technology, Government of India
- Secured an **All India Rank of 577** in Joint Entrance Examination (**JEE Advance 2021**) among 2.5 lakh students enrolled
- Secured an **All India Rank of 677** in Joint Entrance Examination (**JEE Mains 2021**) among 1.2 million students enrolled
- Among **top 250** students qualified to appear for Indian national Astronomy Olympiad 2020-21 among total of 18,000 students
- Among **top 350** students who qualified Indian Olympiad Qualifiers in Physics 2020-21 part 1 with a total of 20,000 candidates

Key Projects

- **What's Next** | *Course Project, Prof. Indranil Saha, IIT Kanpur*  (Jan'23 - May'23)
 - Developed a **web application** using **MERN Stack** which enables event scheduling and delivers personalized notifications
 - Created an interactive interface using **React.js** which has like/dislike functionality, user dashboard and event details page
 - Integrated **MongoDB** as the **backend database** to store event details, enabling efficient retrieval, manipulation and sorting data according to various parameters, while leveraging **Express.js** and **Node.js** to build a secure server-side API
- **Bubble Machine** | *Course Project, Prof. Urbi Chatterjee, IIT Kanpur*  (Jan'23 - May'23)
 - Developed **Verilog processor** with **custom Instruction Set Architecture**, having word and instruction size of 32 bits
 - Designed **FSM** to **fetch instructions** from **memory**, then decode and execute them using **ALU** based on **MIPS-32 ISA**
 - Tested processor for execution in **single clock cycle** using machine code of **Bubble Sort** generated using custom ISA
- **IntelliClassifier** | *Course Project, Prof. Purushottam Kar, IIT Kanpur*  (Jan'23 - May'23)
 - Examined efficacy of **LinearSVC** and **LogisticRegression** by optimizing hyperparameters like loss, tolerance, penalty
 - Trained **decision tree** model to play word guessing game achieving accuracy of **99%** by using **Entropy with Gini Index**
 - Achieved **1.896 MAE** with a training time **1.23s** for KNN after testing various models (**KNN, SVM, Kernel and DNN**) on data set with features like (timestamp, temperature, humidity, sensor readings) for predicting O_3 and NO_2 levels
- **Demystifying Algorithms** | *Science and Technology Council, IIT Kanpur*  (Apr'22 - May'22)
 - Studied about sieve of eratosthenes, meet in the middle technique, binary lifting, bitmasking and dynamic programming
 - Learnt how to tackle **range query problems** using concepts like **prefix sum**, **difference array**, techniques like **square root decomposition**, **Mo's algorithm** and data structures like **sparse tables**, **segment trees** and **Fenwick trees**
- **Unmasking CP** | *Association of Computing Activities, IIT Kanpur*  (Jun'22 - Jul'22)
 - Participated in contests and learnt topics like **binary search**, **linear sieve**, **modular inverse** and **binary representation**
 - Learnt various **Graph Algorithms** like Depth-First Search, Breadth-First Search, Dijkstra's Algorithm, Prim's Algorithm, Kruskal's Algorithm, Bellman ford algorithm, Flyod Warshall Algorithm, Topological Sorting and Disjoint Set (Union-find)

Relevant Courses

Data Structure and Algorithms	Discrete Mathematics	Fundamentals of Computing
Linear algebra and ODE	Introduction to Machine Learning	Software Development and Operations
Probability for Computer Science	Logic in Computer Science	Computer Organization

Technical Skills

- **Programming Languages and Libraries:** C, C++, Python, Numpy, Pandas, Matplotlib, Scikit Learn, Verilog, Qtspim
- **Development and Utilities:** HTML, CSS, JavaScript, Express.js, Node.js, React.js, MongoDB, Git, Bash, Excel, L^AT_EX

Positions of Responsibility

- **Science and Technology Secretary, Hall 6, IIT Kanpur** (Apr'23-present)
 - Assisted in overseeing the preparation of around **200** female students for Takneek, SnT code and Galaxy (Inter Hall events).
 - Worked on automating guest room and mess services, streamlining process of booking guest rooms and ordering in mess.
- **Secretary, Career Development Wing, Academics and Career Council, IIT Kanpur** (Oct'22-May'23)
 - Successfully organized tests, workshops and guidance sessions for UG students to help them prepare for their career paths
 - Contacted professionals in diverse fields to share their expertise as a guest presenter and also to gather sponsorship
- **Secretary, Association of Computing Activities, IIT Kanpur** (Oct'22-May'23)
 - Successfully managed, organized and hosted Happy Hours, Hackathons, Semester Projects and Departmental Farewell